In-Class Activity – Week 10

Part 1

Question 1 – Program of Study and Other Relevant Education

My current program of study is Cybersecurity at Durham College. I use my three-year college diploma in Computer Programming and Analysis at Durham College as a prerequisite to successfully enter the Cybersecurity program at Durham College. Before that, I studied Business Administration at the University of Toronto followed by dropping out in the second year.

Question 2 – Interests and Hobbies

An interest I have is playing video games in competitive environments where my success depends on my strategy and the strategy of my opponents. I enjoy thinking about interactions between defenders and attackers while optimizing and critically thinking about my strategy in attempt to increase my performance. This relates to cybersecurity as game theory can be applied to attackers and defenders in computer systems.

Secondly, a hobby I have is playing Pickleball where anticipation and positioning is important in scoring points, where anticipation of malicious actors in cybersecurity is also vital. Reading the body language of your opponent in Pickleball is similar to understanding motivations, targets, and methods of attackers in Cybersecurity.

Lastly, I am interested in reading patch notes which are documents released by game developers sharing the details of changes, fixes, and additions from an update of the game. Patch notes relate to cybersecurity, specifically in reading vendor security bulletins, as these bulletins reveal common vulnerabilities and exposures while including technical details of fixes.

Question 3 – Favorite Courses in the Cybersecurity Program

Information Security

With the help of the instructor, it has been the most enlightening in terms of learning what the Cybersecurity industry entails.

Cybersecurity Careers and Digital Portfolio

The course greatly helped me understand the type of career path I want and the opportunities available to me after graduating from this program.

Operating System Security

The content and the labs are challenging yet interactive and enjoyable. I use Windows systems on a daily basis and this course helped me learn possible vulnerabilities on these systems.

Wireless Network Security

The content labs are also enjoyable, interesting, and most engaging. I use wireless networks on a daily basis and understanding the tools and techniques in protecting wireless networks helps me learn methods to protect my data while using these wireless networks.

Question 4 – Skills, Strengths, and Experience

Windows Operating System

Uses Windows operating systems consistently. Thoroughly understands features and tools on Windows and knows how resources on such a system is utilized.

Application Programming and Scripting

Background in computer programming and analysis greatly helps with building comprehensive and effective programs using various computer languages such as Python and Java.

Critical Thinking and Problem-Solving

Able to think strategically and creatively, especially under pressure. Effectively creates connections and solutions while breaking down complex programs into manageable components.

Adaptability and Continuous Learning

Willingly embraces and learns new technologies, new standards, new techniques to remain relevant and effective. Able to remain calm and focused under pressure to make effective decisions.

Question 5 – Personality and Work Style

I thrive in high-stake situations. I do not rush to make decisions and am willing to wait for optimal timing or information before executing a plan. I like to remain disciplined and avoid impulsive actions. I enjoy being fully engaged while I am working. I can process information quickly and react effectively. I also enjoy being flexible and can switch to different types of tasks while not being affected by sudden changes in requirements. I also work well in collaborative environments that suggest clear communication and trusting team members in achieving goals.

Question 6 – Values in Future Jobs

Continuous learning and job stability would be the most important to me in a future job. Continuous learning is important for me as it helps me increase my value in the industry and helps me adapt in the constantly changing environment of cybersecurity. Job security and is important for me because it provides financial stability by guaranteeing a reliable income to cover basic needs.

Question 7 – Goals

My goals is to land a job as a Penetration Tester because I can use both my computer programming and analysis skills and cybersecurity knowledge to perform well in such a position.

Part 2

Based on the detailed information you've shared about your education, interests, skills, and goals, here is a comprehensive career guide tailored for you in cybersecurity:

Three Potential Cybersecurity Specializations

Specialization Category	**Suggested Role**	**Why It Fits You**
Red Team (Offensive)	Penetration Tester (Ethical Hacker)	Your goal is to become a Penetration Tester. Your strong background in computer programming and analysis (Python, Java), strategic thinking from competitive gaming, and understanding of attacker-defender dynamics align well with offensive security roles that emphasize finding vulnerabilities and exploiting them ethically.
Blue Team (Defensive)	Incident Responder / Security Analyst	Your interest in anticipation and positioning (similar to Pickleball skill) suits roles requiring proactive defense and incident handling. Your skills with Windows OS and wireless security knowledge complement defensive tasks in monitoring, detection, and response.

Specialization Category	Suggested Role	Why It Fits You
Engineering & Architecture (Build)	Security Engineer	Your programming background and continuous learning mindset equip you well to build and maintain secure systems. Knowledge from operating system security courses and scripting positions you to develop tools and systems that protect infrastructure.

Step-by-Step Roadmap

During Remaining College Time:

- Complete key courses: Information Security, Operating System Security, Wireless Network Security, Cybersecurity Careers and Digital Portfolio.
- Obtain foundational certifications such as CompTIA Security+ and begin working toward ethical hacking certifications like Certified Ethical Hacker (CEH) or OSCP prep.
- Build projects for your portfolio: e.g., small pen testing reports, wireless security assessments, Windows security configuration labs.
- Contribute to open source security tools or participate in Capture The Flag (CTF) challenges online (e.g., TryHackMe, Hack The Box).

First 1-3 Years in Workforce:

- Seek internships or junior roles in penetration testing, security analysis, or security engineering.
- Pursue advanced certifications such as Offensive Security Certified Professional (OSCP) or GIAC Penetration Tester (GPEN).
- Develop a robust digital portfolio documenting your work, labs, and bug bounty or CTF participation.
- Gain experience with industry tools (Metasploit, Burp Suite, Wireshark, SIEM platforms).
- Focus on networking, scripting automation, threat hunting, and incident response.

Technical & Soft Skills to Develop

Technical Skills:

- Using penetration testing tools (Metasploit, Burp Suite, Nmap).
- Writing and automating security scripts in Python and PowerShell.
- Mastery of Windows and Linux operating system security configurations.
- Wireless network security assessment and defense techniques.
- Packet analysis and log monitoring with Wireshark and SIEM tools.

Soft Skills:

- Technical report writing to clearly communicate vulnerabilities and remediation.
- Critical thinking and strategic problem solving, especially under pressure.
- Effective communication for collaboration with teams and presenting findings.
- Adaptability to rapidly changing security environments.
- Discipline and patience in methodical penetration testing and defense.

Actionable Next Steps (Next Month)

- Start the "Pre-Security" or "Intro to Penetration Testing" learning path on TryHackMe to gain structured hands-on practice.
- Set up a home lab using VirtualBox or VMware, installing Windows and Linux VMs for practicing security configurations, exploits, and defense.
- Join a local or virtual cybersecurity meetup, webinar, or Discord community to network with professionals and learn about opportunities.

This plan leverages your strong programming foundation, strategic mindset from gaming and sports, and passions in security knowledge updates to build a successful cybersecurity career—especially toward penetration testing with room for defensive and engineering roles. Continuous learning and hands-on experience will be key to your growth and job stability.

Part 3

Question 1

The suggested cybersecurity role does align with my current career goals and expectations of being a Penetration Tester. The reasoning of the LLM for this role was because I have a background in computer programming and analysis, I have strategic thinking, and I have an understanding in attacker-defender dynamics.

Question 2

One thing that seems the most achievable for me in the next six months would be obtaining foundational certificates such as CompTIA Security+. One thing that seems the most challenging is to seek internships and junior roles as, from my understanding, the job market is highly competitive and can be scarce at times.

Question 3

I can use the actionable next steps to start connecting with real people to get human perspectives on these career paths by talking with people in cybersecurity communities or groups online. From there, I can talk to people with interests in the cybersecurity industry and get information about the right approach with getting involved in cybersecurity.